

UBT502: FOOD PROCESSING

L	T	P	Cr
3	0	0	3.0

Course Objective: To provide basic knowledge about food processing. To acquaint the students with the effects of processing and preservation techniques on physical, chemical and logical properties of food.

Syllabus

Thermal and cold processing: Thermal processing: classification of thermal processing methods, factors affecting thermal resistance of microorganisms and their spores, thermal death time, lethality concept. Refrigeration and cold storage, changes in food during refrigerated storage. Freezing and frozen storage, changes during freezing, freezing methods.

Food dehydration and concentration: Dehydration of foods: normal drying curve, change in food due to drying, drying methods and equipment. Food concentration, methods of concentration, changes in food due to concentration, intermediate moisture foods.

Food irradiation and microwave heating: Irradiation: mechanism of the killing of microorganisms, factors affecting irradiation of foods, effect on the nutritional content of the foods. Microwave heating: working principle, components, food applications, effect on micro-organisms.

Other processing techniques: Smoking, preservatives (natural and artificial), antibiotics usage and side effects. Recent methods in food processing and preservation: pulse electric field, high pressure, ultrasound, ohmic and infrared heating.

Packaging and labelling of foods: Packaging material and their selection, types and levels of packaging, novel packaging techniques. MAP; applications for fresh-prepared produce oxygen, ethylene and CO₂ scavenging technology, zero energy chamber, hypobaric storage.

Course Learning Objectives (CLO)

The students will be able to:

1. Comprehend the working principle and mechanism of action of individual food processing methods on microorganisms and enzyme activity to the increase shelf life.
2. Apply various food processing and preservative techniques for shelf-life extension of foods.
3. Explain the principle and mechanism of processing food using ionic radiations and microwave radiations.
4. Compare the efficiency of different food processing and preservation techniques and know their pros and cons.

Approved in 1st meeting of the Academic Council held on September 11, 2025. Further revision approved in 3rd meeting of the Academic Council held on March 18, 2026, incorporating modification in credits due to the addition of AI courses.

5. Identify different packaging materials used for food packaging and understand their importance.

Text Books

1. Norman N. Potter and Joseph H. Hotchkiss - Food Science, 5th ed. (2012)
2. M. Shafiur Rahman Handbook of Food Processing, 3rd Edition, CRC Press Taylor and Francis group (2020)
3. Frazier William C and Westhoff, Dennis C.)- Food Microbiology, TMH, New Delhi. (2017)
4. Mc Elhatton, Anna, do Amaral Sobral, Paulo José (Eds.) - Novel Technologies in Food Science, Vol.7, 2nd Edition, CRC Press Taylor and Francis group. (2012)

Reference Books

1. Avantina Sharma Textbook of Food Science & Technology (Vol-I & II), International Book Distributing Company, 3rd Ed. (2021)
2. Cheung, Peter C. K., Mehta, Bhavbhuti M. Handbook of Food Chemistry, Springer-Verlag Berlin Heidelberg, 1st Edition. (2015)

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1.	MST	25-35
2.	EST	35-45
3.	Sessionals (May include assignments/quizzes)	30

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